ABB	Smart Winch (+N5800) - Specification Sheet	Revision 2e	Page 1 of 2

- Fill in as much information as possible. Supply drawing or sketch of winch mechanics & dimensions if possible.
- System are generally designed to supply 150% of rated load for intermittent duty during payout & haul. For example, a 100 Ton rated system will manage 150 Tons intermittent duty.
- ABB will work out an optimum system for your application and supply performance curves.

Customer Reference:	**Dated:**

1. Winch Application & Modes of Operation (can select more than one)

☐ General ☐ Tow Escort with Auto Haul / Payout and Tow Push (Constant Tension) ☐ LARS (Launch & Recovery System)

☐ AHC (Active Heave Compensation) ☐ Oceanographic ☐ Umbilical / Tether ☐ Spud winch ☐ Other (specify on page 2)

2. General

Supply Voltage & Hz	
Ambient Temperature Drive Cabinet Location (°C/F)	
Ambient Temperature Motor Location (°C/F)	
Marine Regulations Applicable (DNV, Lloyds, ABS, RINA, BV , other please specify)	

3. Drum Data

Bare Drum = Layer 0

Bare Drum Dia (mm / in)	
Drum Length (mm / in)	
Bare Drum Inner Dia (mm / in)	
Rope Dia (mm / in)	
System Efficiency (%) (Usually 90% or over)	

4a. Gearbox (if applicable)

Gearbox Ratio (ratio)	
Gearbox Location (select one only)	☐ Motor Shaft ☐ Drum Shaft
Can we change this ratio (For optimization)	☐ Yes ☐ No

4b. External Gearing (if applicable)

Gearing additional to the gearbox (eg Pinion & Bull gear)

Small Gear Teeth (teeth) (For gearing ratio calculations)	
Large Gear Teeth (teeth) (For gearing ratio calculations)	
Large Gear Dia (mm / in) (For inertia calculations)	
Large Gear Width (mm / in) (For inertia calculations)	
External Gearing Location (select one only)	☐ Motor Shaft ☐ Drum Shaft
Can we change this ratio (For optimization)	☐ Yes ☐ No

5. Winch Load Specifications

These specifications are used to size the drive & motor for the winch, not for the winch mechanical brake.

Rope Layer at which Specifications Below Apply (Usually Layer 0)	☐ Layer 0 (Bare Drum) or ☐ Specify - Layer Number =
Rated Load (kg / lbs)	
Maximum Continuous Speed at Rated Load (mpm / fpm)	
Light Load (kg / lbs) (Usually light load is 10% to 30% of the rated load)	
Maximum Control Speed at Light Load (mpm / fpm)	
Maximum or Final Layer to be Wound	

6. Calculated or Estimated Motor Size (if available)

- If multiple motors (maximum 10) are used the drive train gearing needs to be same for multiple motors. Describe details in section *9. Comments & Notes*.
- ABB will assist to calculate optimal motor size.

Motors Total Motors	kW / HP Rated Per Motor	RPM Rated	Volts Rated	Hz Rated

7. Options

All options require encoder mounted on Motor or Drum shaft. If encoder mounted on Drum Shaft then confirm with ABB as performance depends on total gearing and encoder resolution.

a	**Motor is used as brake during process. Mechanical brake only used as fail safe or when motor is Off** (Motor needs to be sized to manage the load continuously at zero speed, motor may require forced cooling)	☐ Yes	☐ No
b	**Auto Haul & Payout with Line Tension Limit Set via Potentiometer (usually for Towing application)** (Option **a** necessary to use this feature)	☐ Yes	☐ No
c	**AHC (Active Heave Compensation)** (Option **a** necessary to use this feature)	☐ Yes	☐ No

8. Encoder and Factors Influencing Calculations

- Operation of the Smart Winch system is based on the principle that the electric winch motor is always active during all processes of load winding, unwinding and holding position. The mechanical brake is only used as a failsafe option or when the electric winch motor is not active.
- The system actively monitors the drum turns from the encoder mounted on motor (preferred) or drum. Encoder[1] of type incremental (preferred) or absolute can be used. When incremental encoder is used the system stores the last count in its non-volatile memory. If winch drum or motor is rotated while the Smart Winch system is powered OFF then it is recommended to use Absolute encoder so as to maintain turn count of the system or else Payout Calibrate or Reset Count needs to be done again.
- The encoder turns information is used to compute the effective drum layer and radius which is used to work out rope force or tension, length and linear speed.
- The accuracy of the effective drum layer and radius depends on how well the winch spools the rope symmetrically on the drum.

[1] *Incremental Encoder* needs to be type HTL or TTL, Dual Channel, Differential Signals, 1000ppr, 24VDC supply. Usually 3 x twisted pair = 6 wires used. *Absolute Encoder* needs to be EnDat 2.2 with 5VDC or 24VDC supply and only the Supply, Clock and Data signals need to be used (3 x twisted pair = 6 wires) as the communications is fast enough to not warrant use of any further signals. Confirm with ABB.

Encoder Mount Location All options require encoder mounted on Motor or Drum shaft. If encoder mounted on Drum Shaft then confirm with ABB as performance depends on total gearing and encoder resolution.	☐ Motor Shaft (preferred) ☐ Drum Shaft
Clutch Used Clutch anywhere between drum and motor. If clutch is used then it is preferable to mount the encoder on the drum. If winch drum or motor is rotated while the Smart Winch system is powered OFF then it is recommended to use Absolute encoder so as to maintain turn count of the system or else Payout Calibrate or Reset Count needs to be done again.	☐ No ☐ Yes
Spooling Symmetry How well the rope is wound on the drum at different layers. This effects the accuracy of the system calculations for effective drum layer and radius which is used to work out rope force or tension, length and linear speed.	Spooling Symmetry on a scale of 1 to 10 where 10 = perfect spooling. ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8 ☐ 9 ☐ 10
External Measure Device for Rope Length Smart Winch offers the option of having an extra encoder mounted on an external measure device, such as the sheave pulley, for accurate payout (rope length) measurement in cases where spooling is not symmetrical enough to give required accuracy of rope length measurement.	☐ No ☐ Yes

9. Comments & Notes

Supply drawing or sketch of winch mechanics & dimensions separately if required.